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BSIT 2nd Year

21st Century IT Skills 16-609

AI Tool Prompt Engineering Report

Name of AI Tool

- ChatGPT

Purpose / Domain

The usage of ChatGPT is for its Code assisting functionalities.

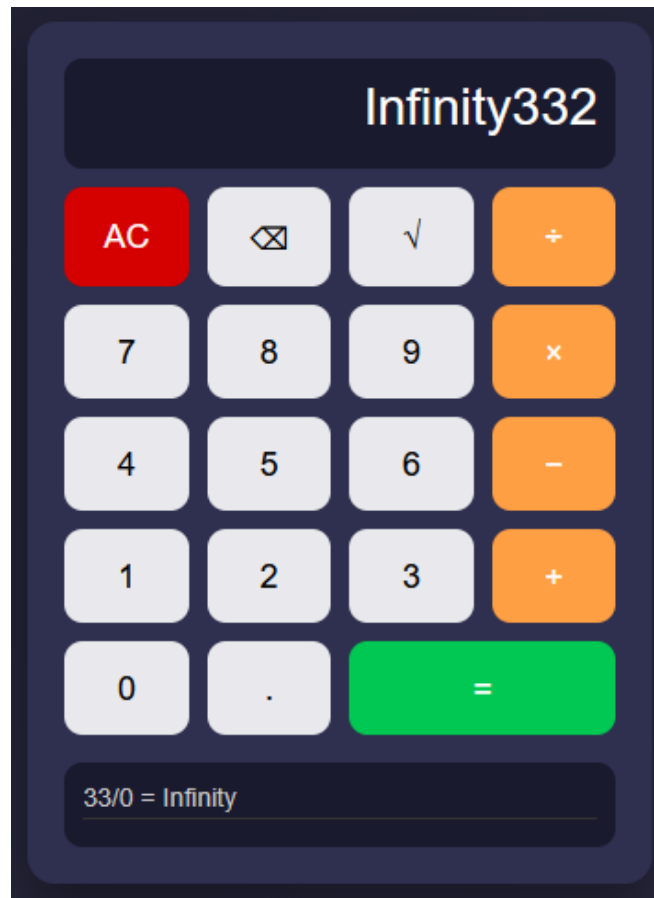
Why I chose

The reason for choosing chatGPT is because they are one of the best ai coding assistant along side with Claude AI. Claude is very reliable based on my experience so I wanted to personally test the capabilities of chatGPT

Original Prompt

Act as a professional programmer with great knowledge of vanilla css, html and javascript. I want you to create me a calculator application website. Everything should be functional, it should be able to add, subtract, divide, multiply, get the square root, there should be numbers from 0-9 and decimals should be allowed. after a calculation the result should be saved, it should also have a backspace button and a delete all button.

Screenshot of Output



LLM Used

- ChatGPT

Improved Prompt

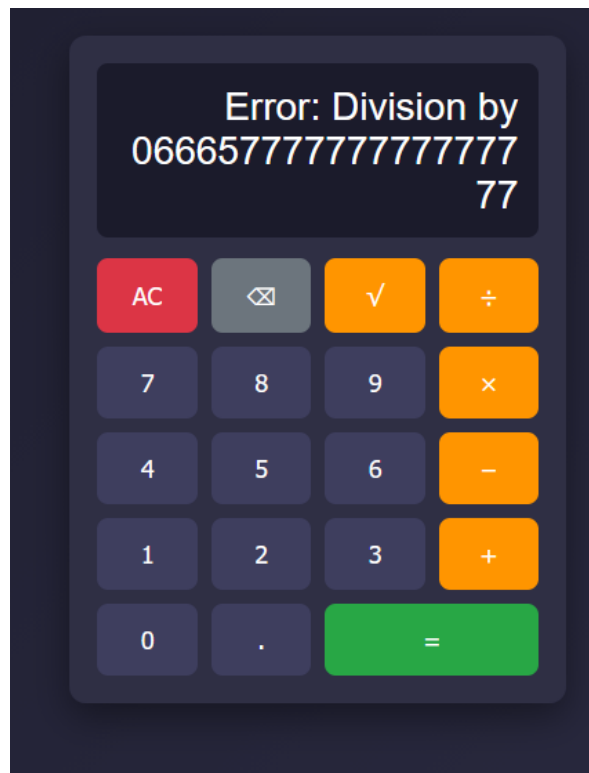
Prompt Used for Improvement:

I want you to improve this prompt so that the AI I will be using will be able to clearly understand what I want to happen. Fix the grammar and add more clarifications in the text (also the final prompt should be in a paragraph form so it is easy to copy and paste):

Result:

Act as a professional front-end developer with strong expertise in vanilla HTML, CSS, and JavaScript (do not use any frameworks or external libraries). Create a fully functional calculator web application that runs directly in a browser. The calculator must support addition, subtraction, multiplication, division, and square root operations. It should include numeric buttons from 0 to 9, allow decimal input (such as 3.14), and correctly handle calculations involving both integers and decimals. The interface should clearly display the current input and the calculated result. After completing a calculation, the result should remain visible and be usable for further operations. Include a Backspace button that removes the last entered character and a Clear (Delete All) button that resets the entire input and display. Ensure all buttons are fully functional and clickable, handle edge cases such as division by zero by displaying an appropriate error message, and write clean, well-structured code with separate sections for HTML, CSS, and JavaScript.

Screenshot of new output



Written Reflection

The two results I received from the AI both have some problems that need to be fixed. In the first version, there is a big issue when dividing by zero. Instead of showing a proper error message, the calculator displays the word "Infinity." When I then press another number, the new number gets added to the word "Infinity" instead of replacing it. This causes the calculator to stop working correctly. The only way to fix it is to reset everything. This should not happen. If there is an error like division by zero, the calculator should clearly show an error message and reset properly when the user starts typing again.

The first version was closer to what I originally asked for in terms of storing the result. I wanted the calculator to keep the result on the screen so that I could continue using it for the next calculation. However, the AI understood this differently. Instead of simply keeping the result available for further operations, it added a history section that stores all previous results. While having a history can be helpful, that was not what I asked for. I only wanted the result to remain on the screen and be usable in the next calculation.

The second version looked better in terms of design. The user interface was cleaner, more modern, and more organized. I personally prefer how it looks compared to the first version. However, it completely ignored my request about storing the result. After getting a result, it did not properly allow me to continue calculating with that same result, which was one of my main requirements.

In short, the first version followed my instructions more closely but had technical problems and poor error handling. The second version looked better but did not fully meet my functional requirements. The best solution would be to combine both: fix the errors, handle division by zero correctly, keep the result usable for the next calculation, and maintain the improved design.